HARRISON MUHORO

Curriculum Vitae · Full-Stack Developer

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Portfolio: harrisonmuhoro.wasmer.app

Date of Birth: 7th March 2006
Nationality: Kenyan
Location: Nyeri, Kenya
Languages: English (Fluent), Kiswahili (Native)

PERSONAL PROFILE

I'm a developer who builds for the Kenya I actually live in — not the one in generic tutorials. My projects exist because I see real gaps: hospitals still running on paper logs, TVET students navigating attachment with no proper system, hotels managing operations across WhatsApp and spreadsheets. Through Cognitara Technologies, I've spent the past year shipping software to close those gaps. My code doesn't just work — it handles authorization properly, isolates data at the ORM layer, and gets security-audited before it touches a public repository. I'm 20 years old and already the kind of developer who patches CVEs, writes canonical decisions documents, and loses sleep over authorization bypasses. I'm looking for an attachment, internship, or full-time role with a team that takes software craft as seriously as I do.

TECHNICAL SKILLS

Backend: Laravel 11/12, PHP 8.3, MySQL, Node.js, Python

Frontend: TypeScript, Blade, React 18, Vite, Tailwind CSS v3

Mobile: Kotlin, Jetpack Compose, Material3 (Android)

Databases: MySQL 8+ / MariaDB (relational), PostgreSQL

DevOps & OS: Git, GitHub, Docker, Linux/Ubuntu, WASM

Integrations: M-Pesa Daraja API, Resend API, Google OAuth (Socialite)

Security: RBAC, OWASP practices, Rate Limiting, Audit Logging, OTP Auth

Other: REST API design, Queue workers, PHPUnit/Pest testing, Responsive UI

WORK EXPERIENCE

IT Support Attachment Student

Nyeri County Government — Department of ICT · September 2026 – November 2026 · Nyeri, Kenya

Industrial Attachment · Second Placement

- Returned for a second placement, continuing to provide IT support across hardware, software, and connectivity issues for department staff
- Assisted with configuration, maintenance, and troubleshooting of ICT infrastructure within the county government
- Built on experience from the first placement to work more independently across support tasks

IT Support Attachment Student

Nyeri County Government — Department of ICT · September 2025 – November 2025 · Nyeri, Kenya

Industrial Attachment · First Placement

- Provided day-to-day IT support to department staff, diagnosing and resolving hardware, software, and connectivity issues
- Configured and set up computer systems and peripherals for county government users across the department
- Assisted with routine maintenance and troubleshooting of ICT infrastructure
- Gained direct exposure to how IT operations run at scale inside a government institution serving Nyeri County

SOFTWARE PROJECTS

Afya HMS — Modular Multi-Branch Hospital Management System

Laravel 12 · React 18/Vite · MySQL · Tailwind CSS · Resend API · Google OAuth (Socialite)

- Designed the full hospital architecture from scratch — multi-branch, modular, built to support complete patient workflows from registration to discharge
- Built the role system entirely: 10 roles from Superadmin to Patient, each with its own permission layer enforced at every API route

- Added Google Sign-In and OTP email verification so patients can register themselves without needing a receptionist to manually create their account
- Found a real authorization bypass on the Patient model — routes with no policy enforcement — and closed it before it could leak data between branches
- Engineered BranchScope as a Laravel global scope to enforce patient data isolation at the ORM level, preventing any cross-branch data leakage
- Upgraded from Laravel 11 to 12 mid-project after CVEs were disclosed — no downtime, no data loss
- Built the public-facing product landing page in Next.js so the HMS platform could have a proper market-facing presence
- Wrote the canonical decisions document: one source of truth for RBAC matrix, queue pipeline rules, and data editing policies across the entire codebase

Taara HMS — Enterprise Hotel Management System

Laravel 11 · PHP 8.3 · TypeScript · Blade · Tailwind CSS v3 · Vite · MySQL · Resend API

- Built an enterprise hotel management system covering the full property lifecycle — reservations, front-desk check-in/check-out, housekeeping, maintenance, billing, and financial reporting in one platform
- Implemented granular RBAC with SuperAdmin, PosPolicy, and StaffPolicy tiers, enforcing strict access control across every operational department
- Built an integrated Restaurant POS module with checkout-to-room billing and direct payment processing
- Designed asynchronous queue workers for PDF and CSV report exports, off-loading heavy processing from the request cycle
- Implemented multi-tier rate limiting: 5 attempts per 15 minutes per account, 20 per IP — with auto-lockouts and Admin manual override endpoints
- Configured production security headers: X-Frame-Options, X-XSS-Protection, HSTS, and Referrer-Policy
- Added full audit logging on all financial mutations with off-hours login tracking and IP tracking on failed attempts
- Covered financial integrity and concurrency edge-cases with PHPUnit and Pest test suites

InduTrack KE — Industrial Attachment Management Platform

Laravel 11 · React 18 · MySQL · GitHub

- An end-to-end platform for Kenya's TVET students — from submitting an attachment application to supervisor sign-off and institutional records
- Three roles, all tested: students applying for placements, company supervisors tracking student progress, and polytechnic admins managing the whole pipeline
- Ran a full security audit before pushing to GitHub: no secrets in the commit history, clean .gitignore, zero environment files exposed
- Secure Laravel backend with authentication, authorization policies, and route protection throughout

Vuka — Industrial Attachment Portal (Swahili: "to cross over")

PHP · MySQL · GitHub

- Named and built the portal — Vuka means 'to cross over' in Swahili, which is exactly what students using it are doing
- Created the brand identity from scratch: Highland Green and Volcanic Clay colour palette with a full visual system
- Deployed clean to GitHub — environment variables for every secret, proper .gitignore, nothing hardcoded anywhere

Personal Portfolio — harrisonmuhoro.wasmer.app

PHP · Resend API · WASM / Wasmer · GitHub

- Built and deployed on Wasmer's WebAssembly runtime — had to figure out dynamic BASE_PATH detection because standard PHP routing assumptions don't hold on WASM
- Working contact form powered by Resend API; no secrets committed to the repository

Unit Converter — Native Android Application

Kotlin · Jetpack Compose · Material3 · Android

- Native Android unit converter in Kotlin and Jetpack Compose — real-time conversion across multiple unit categories with a clean, accessible Material3 interface

Samuel Kiireri Portfolio — Client Developer Portfolio

PHP · MySQL · GitHub

- Designed and developed a professional portfolio site for a client, with dynamic BASE_URL detection for reliable deployment across different environments

EDUCATION

Diploma in Software Development

Nyeri National Polytechnic, Nyeri, Kenya (TVET CDACC Curriculum) · 2024 – Expected March 2028 · In Progress

Key units: Mobile Application Development, Database Application Development, Backend Deployment, Electronics, Communication Skills, Employability Skills

Kenya Certificate of Secondary Education (KCSE)

Mayori Secondary School · 2020 – 2023 · Grade: B−

Kenya Certificate of Primary Education (KCPE)

Kamiu Primary School, Embu · 2015 – 2019 · Mean Grade: B Plain

Primary Education (Class 1–4)

Tenri Primary School · 2012 – 2015

Early Childhood Education

Harvard School, Embu · 2009 – 2011

Baby Class to Top Class

CERTIFICATIONS & TRAINING

Data Analytics Training

Issued by: ICT Authority, Kenya in collaboration with Sigma Digital Technical Institute

Date: 20th July 2026 **Certificate ID:** ICTA-1784540541-7855-46847

Endorsed by the Ministry of Information, Communications & The Digital Economy, Republic of Kenya.

TECH VENTURE

Cognitara Technologies — Founder & Lead Developer

Independent technology venture through which I take on freelance, client, and open-source work. I built it to give my projects a professional home and to hold myself to a higher standard than 'just a student building things'. Focus areas: healthcare tech, hospitality tech, EdTech, and fintech — all targeting the African market.

VOLUNTEER EXPERIENCE

Audio-Visual Technician

Deliverance Church International Skuta · Ongoing

- Manage end-to-end audio-visual operations for live services, crusades, and special events
- Responsibilities: sound engineering and mixing, projection and display systems, live event production coordination

INTERESTS & HOBBIES

Open-Source Development · **African Technology Sovereignty** · **Security Research** · **Gaming** · **Audio-Visual Production**

REFERENCES

Available upon request.